

Bulk and compound-specific stable isotope ratio analysis for authenticity testing of organically grown tomatoes

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ABSTRACT

Until now, there has been a lack of analytical methods that can reliably verify the authenticity of organically grown plants and derived organic food products. In this study, stable isotope ratio analysis of hydrogen (H, $\delta^2\text{H}$), carbon (C, $\delta^{13}\text{C}$), nitrogen (N, $\delta^{15}\text{N}$), oxygen (O, $\delta^{18}\text{O}$) and sulfur (S, $\delta^{34}\text{S}$) was conducted along the tomato passata production process using organic and conventionally grown tomatoes from two Italian regions over two years. A gas chromatography-combustion-isotope ratio mass spectrometry (GC-C-IRMS) based method was developed and applied for analysis of C and N isotope ratios in amino acids derived from tomatoes. Of the bulk isotope ratios, $\delta^{15}\text{N}$ was the most significant parameter for discriminating organic from conventional products. The classification power was improved significantly by compound-specific isotope analysis regardless of the production years and regions. We conclude that isotope analysis of amino acids is a novel analytical tool for complementing existing certification and control procedures in the organic tomato sector.

1. Introduction

Food products, bearing the organic label, should be produced without synthetic fertilizers and pesticides, hormones, genetic engineering, sewage sludge, or irradiation (Council Regulation (EC) No. 834/2007). Organic farmers, processors and traders, must comply with these regulations, if they want to use the EU organic logo and sell their products as organic. In 2008 the European Council of Agricultural Ministers agreed on a new Council Regulation (EC No 967/2008) setting out the principles, aims and rules for organic production as well as defining how organic products can be labelled was instituted. Despite the strict EU regulations for food and food products to be classified as organic, there has been a lack of analytical methods that can reliably verify the authenticity of organic produce (Hong et al., 2017; Kahl et al., 2014; Kelly & Mihailova, 2017; Laursen, Schjoerring, Kelly, & Husted, 2014).

Tomatoes are one of the most frequently consumed horticultural products worldwide with 182 million tons of tomatoes grown globally

in 2017 (FAOSTAT data, accessed December 2019). Italy is the second main producer of tomatoes in Europe with 6 million tons produced in 2017 (FAOSTAT data, accessed December 2019). Italian production of organic tomatoes has almost doubled in two years, with a production of 162,619 tons of organic tomatoes in 2017 (http://www.tomatonews.com/en/italy-the-production-of-organic-tomatoes-has-almost-doubled-in-two-years_2_219.html).

The choice of fertilizer and other agricultural management practices are critical in growing tomatoes and will differ depending on whether the tomatoes are grown organically or conventionally. The prohibition of synthetic fertilizers in organic production does potentially impact the availability of nitrogen including the ratio of $^{15}\text{N}/^{14}\text{N}$, expressed as $\delta^{15}\text{N}$ (Masclaux-Daubresse et al., 2010; Styring, Fraser, Bogaard, & Evershed, 2014a). Synthetic fertilizers have $\delta^{15}\text{N}$ values from -6 to $+6\text{‰}$, which is lower than those of organic fertilizers, ranging between $+0.6$ and $+36.7\text{‰}$ (manure between $+10$ and $+25\text{‰}$) (Bateman & Kelly, 2007). Plants grown conventionally with synthetic fertilizers therefore often have lower $\delta^{15}\text{N}$ values than those grown organically

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(Laursen et al., 2014; Nakano & Uehara, 2007). The roots of non-N₂-fixing plants take up nitrogen from soil as either ammonium (NH₄⁺) or nitrate (NO₃⁻) and the $\delta^{15}\text{N}$ value of the fertilizer, synthetic or organic, will be reflected in the plant (Bateman & Kelly, 2007). Assimilation of NO₃⁻ and NH₄⁺ in plants occurs differently. Indeed, NH₄⁺, once taken up by the roots, is metabolized by glutamine synthetase and enters directly in the amino acids metabolism of the plant. On the other hand, NO₃⁻ is sequentially reduced by nitrate and nitrite reductase enzymes to nitrite and then ammonia (Lea & Azevedo, 2017). Through the assimilation of ammonia, glutamine synthetase produces the first amino acid, glutamine, and plays a central role in the amino acids metabolism. Due to the transfer of the glutamine amide to keto acids and transamination, plants produce glutamate and then other amino acids (Wada, 1995). However, the impact of nitrogen assimilation on stable isotope signatures of plant-derived amino acids is still poorly understood.

Several studies have shown that bulk tissue analysis of stable isotope ratios of N, sometimes in combination with those of C, O, H and S, together with statistical methods, has the potential to differentiate vegetables and fruits grown under different farming systems (Bateman & Kelly, 2007; Camin et al., 2011; Cristea, Feher, Magdas, Voica, & Puscas, 2017; Flores, López, Fenoll, Hellín, & Kelly, 2013; Georgi, Voerkelius, Rossmann, Graßmann, & Schnitzler, 2005; Mahne Opatić, Opatić, Nečemer, Kocman, & Lojen, 2017; Rapisarda et al., 2010; Schmidt et al., 2005; Wada, 1995). However, it has been shown that compound-specific isotope ratio analysis is more powerful (Laursen et al., 2013; Mihailova, Pedentchouk, & Kelly, 2014; Novak, Adler, Husted, Fromberg, & Laursen, 2019; Paolini, Ziller, Laursen, Husted, & Camin, 2015; van Leeuwen, Prenzler, Ryan, & Camin, 2014). The application of bulk tissue stable isotope ratio analysis of N is challenged by the fact that N₂-fixing-plants, used as cover crops or green manures in organic crop rotations, have $\delta^{15}\text{N}$ values overlapping with those of conventionally grown plants due to the atmospheric origin of N in both cases (Laursen et al., 2013; Paolini et al., 2015). This can be by-passed by stable isotope ratio analysis of N and O from nitrate or of O from sulfate, which has proven to be promising for authentication of organic vegetables (Laursen et al., 2013; Mihailova et al., 2014; Novak et al., 2019). Another approach is GC-C-IRMS based isotope ratio analysis of amino acids, which has already proven valuable for discriminating organic and conventional cereals (Paolini et al., 2015).

In the present study, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, $\delta^{34}\text{S}$, $\delta^2\text{H}$ and $\delta^{18}\text{O}$ values were measured along the production chain of tomato passata (soil – tomato – passata), from tomatoes grown organically and conventionally in two regions of Italy. Furthermore, $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ were determined in amino acids extracted from a set of organic and conventional tomatoes from Italy and Denmark. The overall aim of the study was to identify the most suitable isotope parameters for authenticity testing of organically grown tomatoes.

2. Materials and methods

2.1. Standards and reagents

The standards L-Alanine ($\geq 98\%$); Glycine ($\geq 99\%$); L-Valine ($\geq 98\%$); L-Isoleucine ($\geq 98\%$); L-Leucine ($\geq 98\%$); L-nor Leucine ($\geq 98\%$); L-Proline ($\geq 99\%$); L-Aspartic acid ($\geq 98\%$); L-Glutamic acid ($\geq 99\%$); L-Glutamine ($\geq 99\%$); L-Asparagine ($\geq 98\%$); L-Phenylalanine ($\geq 98\%$); L-Threonine ($\geq 98\%$) and the analytical grade Amberlite® IR120 cation-exchange resin (hydrogen form) were purchased from Sigma-Aldrich (Milan, Italy). Triethylamine ($\geq 99.5\%$) and acetic anhydride ($\geq 99\%$) reagents were purchased from Sigma-Aldrich (Milan, Italy). Analytical grade acetone, isopropanol, diethyl ether, dichloromethane and ethyl acetate were purchased from VWR (Milan, Italy) and Sigma-Aldrich (Milan, Italy). Petroleum ether (40–60%) was purchased from VWR Chemicals (Milan, Italy).

2.2. Tomato samples and growth conditions

Samples of tomato (*Solanum lycopersicum* L.) were obtained directly from farms (on-farm trials) during 2012 and 2013, in two different Italian regions, viz. Emilia Romagna (in Central Italy) and Basilicata (in Southern Italy). In both regions, the tomatoes were grown under two different systems: organic (Org) or conventional (Conv). Tomatoes grown within the organic regimen were managed in compliance with the EU regulations for organic farming (European Community Council Regulation EC 834/2007) and had been organic for more than 10 years prior to 2012. The systems were all based on stockless crop production, but were managed differently from each other, due to differences in climate, soil types and the structure of the farm. In the organic production, organic fertilizers and mechanical weed management were applied, and in the conventional production a combination of organic and mineral fertilizers was used as well as chemical weed protection agents. Irrigation was implemented on both the organic and conventional fields. At each location and with both systems, 2 varieties were grown; Roma (long tomato) and San Marzano (round tomato). The tomatoes were planted at the end of April/beginning of May each year and were harvested at maturity from mid-August to early September depending on the region. The total number of tomato samples was 48 (2 locations, 2 years, 2 systems, 2 varieties and 3 replicate sampling times).

Organically grown tomato plants were grown in large plots of a minimum of 1000 m². Tomatoes were sampled on three separate occasions whereby all tomatoes that were ripe on the day of collection were harvested in a large batch and then a small sub-sample of 2 kg was selected for analysis. Due to extreme drought in 2012 a plot in Emilia Romagna was lost and could not be sampled. Harvesting quantities were 20–30 tonnes/ha in the organic system compared to 50–80 tonnes/ha in the conventional one in Basilicata. In Emilia Romagna, 50–60 tonnes/ha was harvested in the organic system and 70–90 tonnes/ha in the conventional system. A soil sample from each region and growth system was also collected (8 in total). The soil in Basilicata was loamy, whereas the soil in Emilia Romagna consisted of clay. Details regarding tomato sample information like varieties, farming system details, climatic data and fertilizers application and use are shown in Tables 1–4 of the Supplementary Material.

Fresh tomatoes were stored at 10 °C for a maximum of 5 days and were then transported to a processing plant in refrigerated vehicles. The processing from tomatoes to passata was only physical, through a ‘cold-break treatment’ (Hayes, Smith, & Morris, 1998). Briefly, tomatoes were cut into pieces and were then heated with salt at a temperature < 60 °C. The tomato flesh was separated mechanically from peels and seeds using a pulper with a hole size of around 1.0–1.3 mm, before being vacuum-concentrated to around 8° Brix. Passata was then bottled and pasteurized. This type of treatment preserves the bright red color, fragrance and full taste of fresh harvested tomatoes. In Basilicata, 4 passata were produced for both organic and conventional systems, whereas 2 passata per agricultural system were sampled in Emilia Romagna (12 samples in total).

In order to test the applicability of the developed analytical method for organic authenticity testing across countries and production methods, a test sample set consisting of 13 different greenhouse grown tomatoes was collected from Danish growers. Of these samples, 8 were organic and 5 were conventional and included a wide range of tomato varieties (classic, plum, snack, cocktail, cherry or San Marzano) and different genotypes, growth medias and fertilization strategies. Organic tomato samples were grown either in soil or in peat mixture. In contrast, conventional samples were grown in artificial growth media (rock wool) or fibers. Organic samples were fertilized with animal manure, garden waste or vegetable-based liquid fertilizer mixture, whereas conventional samples were fertilized with liquid inorganic fertilizers.

All tomato and passata samples were subject to H, O, N, C and S stable isotope ratios analysis on bulk tissue. A number of samples of

Italian tomatoes ($N = 16$), randomly selected from the complete set of samples in order to equally cover the agricultural systems, regions and years, were also used for C and N stable isotope ratio analysis of amino acids. In addition, the Danish tomatoes test set was analyzed for C and N stable isotope ratios of amino acids. All the soils from field-grown tomatoes were analyzed for bulk N and C stable isotope ratios.

2.3. Sample preparation and bulk analysis

Approximately 2 kg of mature tomato fruits were taken for each sample. Samples were rinsed in ultrapure water (Milli-Q Element, Millipore), frozen and freeze-dried (SvanVac Coolsafe 95–15 PRO) at 1.0 mbar until complete dryness. Samples were crushed in a plastic bag followed by milling into a fine powder using a 200 μm mesh size titanium sieve and a titanium rotor in a centrifugal mill (ZM 200 Retsch GmbH, Haan, Germany) and stored in plastic bags at room temperature until analysis. The soil was air-dried (at room temperature) and sieved (< 2 mm), while passata samples were freeze-dried and then ground into fine powder.

Tomato and passata samples were measured for stable isotope ratios of C, N, S, H and O. The bulk sample (0.5 mg) was weighed into tin capsules, and was combusted in an elemental analyser (VARIO CUBE, Elementar Analysensysteme GmbH, Germany) coupled to an isotope ratio mass spectrometer (IRMS, Isoprime, Isoprime Ltd., UK) to measure the $^{13}\text{C}/^{12}\text{C}$, $^{15}\text{N}/^{14}\text{N}$ and $^{34}\text{S}/^{32}\text{S}$ isotopic ratios in one run. In the case of soils, $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ were determined using an elemental analyser (Flash EA 1112, Thermo Scientific, Bremen, Germany), interfaced through a ConFlo IV dilutor device (Thermo Finnigan, Bremen, Germany) with a DELTA V Thermo Scientific isotope ratio mass spectrometer. A pyrolyser (Finnigan TC/EA, high temperature conversion elemental analyser, Thermo Scientific) coupled to an IRMS (Finnigan DELTA XP, Thermo Scientific) was used to measure the $^2\text{H}/^1\text{H}$ and $^{18}\text{O}/^{16}\text{O}$ isotopic ratios in one run for the bulk tomato and passata sample (0.5 mg) in silver capsules. Samples were measured in duplicate. The isotope ratios were expressed in ‰ against Vienna Standard Mean Ocean Water for $\delta^{18}\text{O}$ and $\delta^2\text{H}$, Vienna-Pee Dee Belemnite for $\delta^{13}\text{C}$, air for $\delta^{15}\text{N}$ and Vienna Canyon Diablo Troilite for $\delta^{34}\text{S}$ according to the following equation:

$$\delta_i E = \frac{i R_{SA} - i R_{REF}}{i R_{REF}} \quad (1)$$

where R_{SA} is the isotope ratio measured for the sample and R_{REF} is the international standard isotope ratio. The isotopic values for $\delta^{13}\text{C}$, $\delta^{15}\text{N}$ and $\delta^{34}\text{S}$ were calculated against working in-house standards, which were themselves calibrated against international reference materials: potassium nitrate IAEA-NO3 (IAEA-International Atomic Energy Agency, Vienna, Austria) for $\delta^{15}\text{N}$, L-glutamic acid USGS 40 (U.S. Geological Survey, Reston, VA, USA) for $^{13}\text{C}/^{12}\text{C}$ and $^{15}\text{N}/^{14}\text{N}$, fuel oil NBS-22, IAEA-CH-6 for $^{13}\text{C}/^{12}\text{C}$, barium sulphates IAEASO-5, NBS 127 (IAEA) and USGS 42 and USGS 43 for $\delta^{34}\text{S}$. Through the development of a linear equation, the values were calculated against two working standards. The isotopic values for $\delta^2\text{H}$ and $\delta^{18}\text{O}$ were calculated against KHS (Kudu Horn Standard, $\delta^2\text{H} = -35.3 \pm 1.1\text{‰}$ and $\delta^{18}\text{O} = +20.3 \pm 0.3\text{‰}$) and CBS (Caribou Hoof Standard $\delta^2\text{H} = -157 \pm 0.9\text{‰}$ and $\delta^{18}\text{O} = +3.8 \pm 0.3\text{‰}$) through the development of a linear equation and implementing the comparative equilibration procedure (Dunn & Carter, 2018). These keratinous standards were used because of the absence of any international and matrix matched organic reference material. For the $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, $\delta^{34}\text{S}$, $\delta^{18}\text{O}$ and $\delta^2\text{H}$ the uncertainty of measurement (2 Std Dev) was 0.1‰, 0.2‰, 0.3‰, 0.3‰ and 1‰, respectively.

2.4. Preparation and GC-C-IRMS analysis of amino acids from tomatoes

Extraction, hydrolysis, purification and derivatization of amino acids from tomatoes was carried out as described for wheat by Paolini

et al. (2015). Protein was hydrolyzed with HCl, which was subsequently derivatized to obtain the *N*-acetyl-isopropyl esters of the amino acids. GC-C-IRMS analysis determined the isotopic ratios and retention times for derivatized individual amino acids alanine (Ala), glutamate (Glx), glycine (Gly), isoleucine (Ile), leucine (Leu), phenylalanine (Phe), proline (Pro), threonine (Thr), and valine (Val). The amino acid glutamine was converted to glutamic acid, due to acid hydrolysis. Accordingly, the isotopic values of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for Glx reflect the isotopic values for carbon and nitrogen for both glutamate + glutamine (Paolini et al., 2015; Styring et al., 2012; Styring, Fraser, Bogaard, & Evershed, 2014b). The analysis was carried out according to the method set up by Paolini et al. (2015), with a Trace GC Ultra (GC Isolink + ConFlo IV, Thermo Scientific) furnished with an autosampler (Triplus, Thermo Scientific) and for $\delta^{15}\text{N}$ analysis a HP-INNOWAX column (60 m $\times$ 0.32 mm i.d. $\times$ 0.25 μm film thickness, Agilent). Instrumental performance was monitored using a standard mixture of derivatized L-amino acids with the addition of an internal standard, L-norleucine (Nle), and their isotopic values for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were measured during each analytical run as recommended by Yarnes and Herszage (Paolini et al., 2015; Yarnes & Herszage, 2017). Because of its absence in tomatoes, Nle was chosen as the internal standard for the samples and standard mix and its isotopic value was measured in each sample run. The L-amino acid EA-IRMS isotopic values determined for Nle were -27.6‰ for $\delta^{13}\text{C}$ and $+14.0\text{‰}$ for $\delta^{15}\text{N}$. When the difference between the values determined by EA-IRMS and GC-C-IRMS for the Nle for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were no more than 1.5‰ and 1.0‰, respectively, the analytical run was accepted.

2.5. Data analysis and corrections

The derivatized amino acid $\delta^{13}\text{C}$ values are a composite of the initial starting amino acid carbon and of the reagents implemented in the derivatization reaction. Due to the addition of these extra carbons, an empirical correction was applied to determine the $\delta^{13}\text{C}$ value of the amino acid (Docherty, Jones, & Evershed, 2001; Paolini et al., 2015).

The $\delta^{13}\text{C}$ values measured by EA-IRMS for the non-derivatized amino acid standards and by GC-C-IRMS for the derivatized amino acid standards were utilized to calculate the correction factors:

$$n_{cd} \delta^{13}\text{C}_{cd} = n_c \delta^{13}\text{C}_c + n_d \delta^{13}\text{C}_d \quad (2)$$

where n is the number of moles of carbon, and the subscripts c, d, and cd represent the compounds of interest, the derivative group, and the derivatized compound, respectively. Measurement errors as well as the total analytical error, which arises from the different derivatization steps, were considered when calculating the $\delta^{13}\text{C}$ measurement of uncertainty:

$$\sigma_c^2 = \sigma_s^2 \left(\frac{n_s}{n_c} \right)^2 + \sigma_{sd}^2 \left(\frac{n_s + n_d}{n_c} \right)^2 + \sigma_{cd}^2 \left(\frac{n_c + n_d}{n_c} \right)^2 \quad (3)$$

where n is the number of moles of carbon, and the subscripts c, s, d, cd and sd represent the non-derivatized compound, the non-derivatized standard, the derivatizing agent, the derivatized compound and the derivatized standard, respectively.

Accuracy and precision of GC-C-IRMS analysis was the same as reported in Paolini et al. (2015). In particular, precision was on average $\pm 0.4\text{‰}$ for $\delta^{15}\text{N}$ and $\pm 0.8\text{‰}$ for $\delta^{13}\text{C}$ (1 Std Dev) and uncertainty was on average $\pm 0.3\text{‰}$ for $\delta^{15}\text{N}$ and $\pm 1.0\text{‰}$ for $\delta^{13}\text{C}$ (1 Std Dev).

2.6. Statistical analysis

The data were analyzed with Statistica (version 9, StatSoft Inc., Tulsa, OK, USA). To examine trends for different ratios along the production chain, a paired *t*-test was applied. Analysis of Variance (ANOVA) and Tukey test (HSD or for unequal number of samples) were

applied to test differences related to agricultural system, production year, region and type of sample (soil, tomato, passata). Kolmogorov-Smirnov and Shapiro-Wilk tests were used to check if data were normally distributed.

Glutamine and glutamic acid isotopic values are central to all amino acid isotopic values as all amino acids are derived from these. Single amino acid isotopic values were therefore normalized against corresponding Glx values to control differences associated with peripheral sources of nitrogen, such as locality (Paolini et al., 2015; Styling et al., 2014b).

Principal Component Analysis (PCA) was performed on bulk data for $\delta^2\text{H}$, $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$ and $\delta^{34}\text{S}$ in tomatoes and on data for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of the amino acids from tomatoes. PCA allows multi-dimensional data to be projected into low dimensional subspaces, highlighting the direction of maximum variability in the original dataset. PCA results are presented in graphical form as score and loading plots. In the score plot, each point represents the position of each sample in the isotopic space. In contrast, the loading plot shows the contributions of each isotope ratio to the principal components. All PCA analyses were performed on mean centered and scaled datasets. Finally, a full cross-validation was applied to test the model.

3. Results and discussion

3.1. Trends of bulk tissue isotope values along the tomato production chain

To determine the behavior of the five isotope ratios during passata production for the two different production systems, eight production chains (from soil to tomatoes to passata), randomly selected, were considered (4 from Emilia Romagna, 4 from Basilicata, 4 from organic systems and 4 from conventional systems) (Fig. 1). A paired *t*-test was applied to check statistically significant differences among the matrices collected along the production chain (soil versus tomato and versus passata, tomatoes versus passata) for the five isotope ratios. The differences between the tomatoes and passata were not statistically different according to the paired *t*-test ($p > 0.05$), even when considering separately the organic and conventional production chains. The only exception was $\delta^{13}\text{C}$ between tomatoes and passata when considering only organic samples or all the samples together ($p < 0.01$). It is worth noting that $\delta^{13}\text{C}$ of passata presented higher values ($p < 0.001$) compared to tomatoes, and also the Tukey test produced the same results (Table 1). This can be explained by the loss of peel and seeds during the preparation of passata. In a previous study, apple peels were reported to have more depleted values compared to pulp (Mimmo et al., 2015).

Except for $\delta^{13}\text{C}$, comparable results were found in a previous study (Bontempo et al., 2011; Mimmo et al., 2015), where statistically significant differences were not observed in the processing from tomatoes to juice and finally to passata for the five isotope ratios. From these results, it was evident that the production of passata did not alter the isotopic content systematically in either of the two agricultural systems. This was also the case for oxygen and hydrogen isotope ratios, despite potential evaporation during the production. This could be explained by the fact that during the 'cold-break' process the heating of the chopped tomatoes was carried out at low temperatures (lower than 60 °C) (Hayes et al., 1998). Significant differences were found for $\delta^{13}\text{C}$ values between tomatoes and soils ($p < 0.001$, Table 1). In particular, the $\delta^{13}\text{C}$ values of the soils were higher with respect to the corresponding tomatoes, with a mean difference of 2‰. Similar results were previously observed in another study on apples, pears and grapes (Dehelean, Magdas, & Cristea, 2013). The C of plant materials is derived from atmospheric CO_2 and not from the soil, which may explain the lack of relationship. Carbon isotope ratios of soil are instead expected to be related to the input of organic matter from previous growth years. For $\delta^{15}\text{N}$, no statistically significant differences were observed between soils and tomatoes. This close relationship was expected as N in plants is

derived from plant available N in the soil during the growth season.

3.2. Bulk data in the different matrices

In Table 1 the means and standard deviations of $\delta^2\text{H}$, $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$ and $\delta^{34}\text{S}$ values in soils, tomatoes and passata are presented according to the origin and the agricultural system. The impact of product type (soil, tomatoes and passata), production year (2012 versus 2013), tomato variety (long versus round), geographic origin and conventional versus organic products was statistically evaluated. No statistically significant differences ($p > 0.05$) were found according to the variety or the production year for all the isotope ratios. ANOVA test found statistically significant differences ($p < 0.001$) according to the agricultural system and the geographic origin.

Considering the geographical origin, significant differences were found for $\delta^{18}\text{O}$ and $\delta^{34}\text{S}$ in tomatoes ($p < 0.001$) and $\delta^{34}\text{S}$ of passata ($p < 0.001$) (Table 1). In particular, higher $\delta^{18}\text{O}$ values were found in Basilicata tomatoes than Emilia Romagna samples (mean values and standard deviations $+26.7\text{‰} \pm 0.9\text{‰}$ versus $+24.2\text{‰} \pm 1.2\text{‰}$), as expected due to the different latitudes and climatic conditions of the two regions. Furthermore, lower $\delta^{34}\text{S}$ values were found in Basilicata tomatoes and passata (mean values and standard deviations $-3.0\text{‰} \pm 1.5\text{‰}$ and $-4.4\text{‰} \pm 0.2\text{‰}$) than for the Emilia Romagna samples ($-0.3\text{‰} \pm 2.9\text{‰}$ and $-0.3\text{‰} \pm 1.9\text{‰}$) which were likely related to the different geological characteristics of the soils, differences in the microbial activity as well as the different proximities to the sea (Krouse & Grinenko, 1991).

3.3. Discrimination between organic and conventional tomatoes using bulk isotope data

Statistically significant differences ($p < 0.001$) were found for $\delta^{15}\text{N}$ in tomatoes (Table 1). As expected, tomatoes grown conventionally had lower values of $\delta^{15}\text{N}$ compared to organic samples (mean values $4.5 \pm 2.1\text{‰}$ versus $9.2 \pm 3.3\text{‰}$), irrespective of geographic origin. The $\delta^{15}\text{N}_{\text{bulk}}$ values of the tomatoes ranged between $+4.0\text{‰}$ to $+14.6\text{‰}$ for the organic system and $+1.6\text{‰}$ to $+8.2\text{‰}$ for the conventional system, respectively, and these values depict the nature of the fertilization or farming systems implemented. Fertilizers synthetically obtained via the Haber-Bosch process, have nitrogen derived from the air and therefore have $\delta^{15}\text{N}$ values between -6‰ and $+6\text{‰}$. On the other hand, organic fertilizers derived from animal waste or vegetable matter, have isotopic values of $\delta^{15}\text{N}$ ranging from $+0.6\text{‰}$ to $+36.7\text{‰}$ (Bateman & Kelly, 2007). In this study, farmyard manure was e.g., used for the organic systems which, according to Bateman and Kelly (2007), has $\delta^{15}\text{N}$ values ranging from $+3.5\text{‰}$ to $+16\text{‰}$, whereas KNO_3 used in the conventional systems have much lower values of $\delta^{15}\text{N}$ (from -1‰ to -1.5‰ according to Bateman and Kelly) (Tables 2 and 3 Supplementary Material).

The $\delta^{13}\text{C}$ mean values for the organic and conventional systems were not significantly different in this study (Table 1). The $\delta^{13}\text{C}_{\text{bulk}}$ values of the tomatoes (-29.9‰ to -26.0‰) were representative of C3 plants (-34‰ to -24‰) (Krueger & Reesman, 1982), as they are mainly influenced by the plant's photosynthetic pathway. However, nutrient availability, soil types, CO_2 concentrations, and water availability can also affect the plant $\delta^{13}\text{C}$ values (Longobardi et al., 2017). Moreover, these values relate to the uptake of CO_2 from the atmosphere and therefore to the factors that have an impact on stomatal aperture and photosynthesis, such as temperature, salinity and light intensity (O'Leary, 1981). The use of different farming systems, organic and conventional, could lead to different $\delta^{13}\text{C}$ values, as usually under conventional conditions the availability of nitrogen is high and leads to lower $\delta^{13}\text{C}$ values, whereas lower nitrogen availability leads to higher $\delta^{13}\text{C}$ values, which could help to differentiate between the systems (Paolini et al., 2015).

The $\delta^2\text{H}_{\text{bulk}}$ values of the tomatoes (-53‰ to -24‰) and the

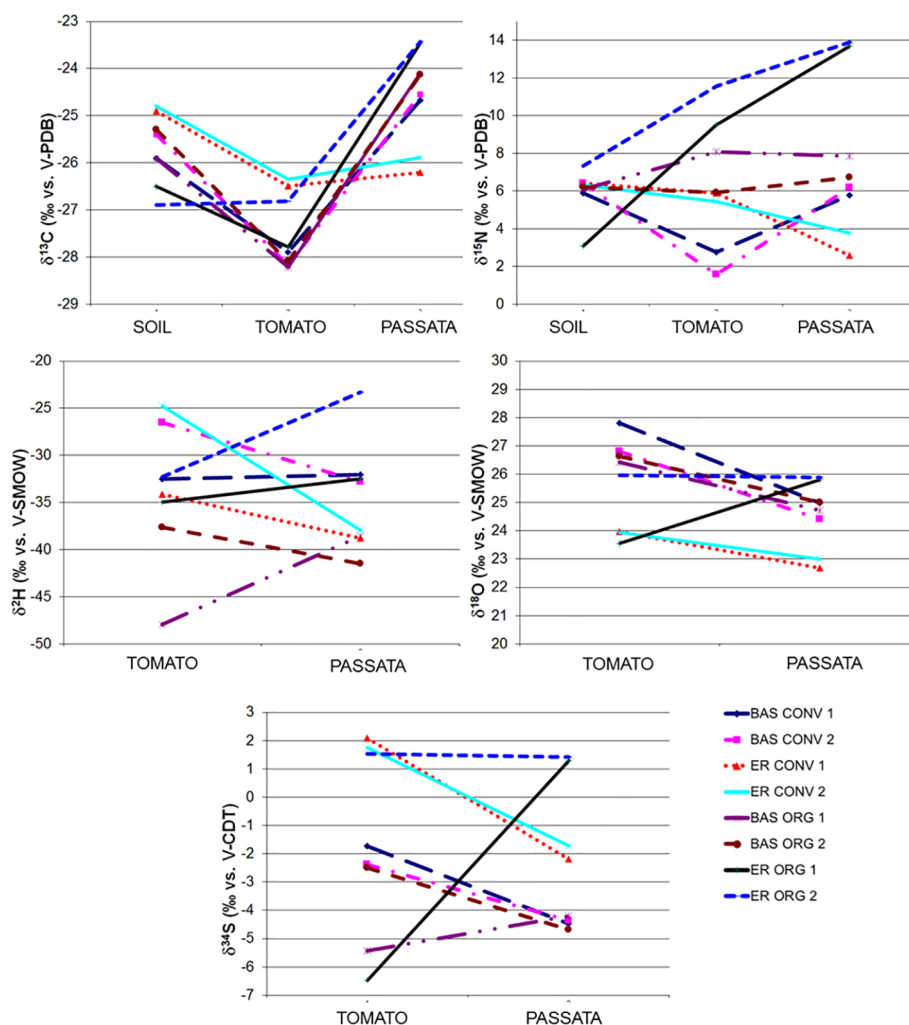


Fig. 1. Trend of stable isotope ratios of H, C, N, O and S determined along the production chain of tomato passata in soils, tomatoes and passata collected in 2012 and 2013 in two Italian regions (Basilicata and Emilia Romagna).

$\delta^{18}\text{O}_{\text{bulk}}$ values ($+22.5 \pm 28.2\text{‰}$) are determined by a number of factors such as temperature, precipitation and humidity, longitude, latitude, and altitude. There were no significant differences between the conventional and the organic system for $\delta^2\text{H}_{\text{bulk}}$ (mean values and standard deviations $-37\text{‰} \pm 8\text{‰}$ for both) and $\delta^{18}\text{O}_{\text{bulk}}$ (mean values and standard deviations $25.0\text{‰} \pm 1.8\text{‰}$ and $25.9\text{‰} \pm 1.3\text{‰}$), which means that the farming system (organic and conventional) in this specific case did not influence these isotopic values. The $\delta^{34}\text{S}_{\text{bulk}}$ values of the tomatoes were between -6.6‰ and $+2.1\text{‰}$ and did not allow discrimination between conventional and organic tomatoes ($-1.7\text{‰} \pm 2.3\text{‰}$ for conventional samples and $-1.7\text{‰} \pm 3.0\text{‰}$ for organic ones).

Principal component analysis was conducted using the bulk data for $\delta^2\text{H}$, $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$ and $\delta^{34}\text{S}$ in tomatoes (Fig. 1 Supplementary Material). The majority of the data variability was explained by the first three PCs with 40% variance explained for the first PC, 27% for the second PC and 19% for the third PC. The first PC had three major contributors, $\delta^{13}\text{C}$, $\delta^{34}\text{S}$ and $\delta^2\text{H}$, whereas $\delta^{15}\text{N}$ mainly contributed to the second PC. No clear grouping according to the farming system was seen (Fig. 1 Supplementary Material) when plotting the first two PCs.

Collectively, results from the multi-element bulk tissue isotope ratio analysis did not provide sufficient classification power to discriminate organically and conventionally grown tomatoes. Of the isotope ratios analyzed, $\delta^{15}\text{N}$ was the most promising marker as it could reveal the use of organic manure. This is contrary to the results of Paolini et al. (2015)

who found that $\delta^{15}\text{N}_{\text{bulk}}$ could differentiate organically and conventionally grown cereal grains. However, $\delta^{15}\text{N}$ has showed clear limitations for organic authenticity testing when organic manures are used in conventional production or when legume based green manures are used in organic production (Laursen et al., 2013; Paolini et al., 2015).

3.4. Discrimination of organic and conventional tomatoes using amino acid isotope analysis

Amino acid isotope ratio analysis has previously eliminated the limitations of bulk tissue isotope ratio analysis in cereal grains (Paolini et al., 2015) and was therefore further explored for authentication of organic tomatoes in the present study. Amino acids (aa), Ala, Val, Ileu, Leu, Gly, Pro, Thr, Glx and Phe, in tomato samples were analyzed for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ on a subset of 16 samples randomly selected in order to equally cover the agricultural systems, regions and harvest years. In Table 2, the means of the $\delta^{13}\text{C}_{\text{aa}}$ and $\delta^{15}\text{N}_{\text{aa}}$ values for the different farming systems, regions and years are reported. A Tukey test showed significant differences ($p < 0.001$) between the organic and conventional farming systems for $\delta^{15}\text{N}_{\text{Ileu}}$, $\delta^{15}\text{N}_{\text{Glx}}$ and $\delta^{15}\text{N}_{\text{Phe}}$. For $\delta^{13}\text{C}$ there was only a significant difference ($p < 0.001$) between systems for Glx. There were no significant effects for $\delta^{15}\text{N}$ or $\delta^{13}\text{C}$ for all amino acids between the different years, 2012 and 2013, and the regions.

The $\delta^{15}\text{N}$ amino acid values were normalized against the $\delta^{15}\text{N}_{\text{Glx}}$ (Fig. 2 Supplementary Material) (Paolini et al., 2015; Styring et al.,

Table 1

Isotopic mean values and standard deviations for soils, tomatoes and passata bulk samples collected in 2012 and 2013 harvest years in two Italian regions (Basilicata and Emilia Romagna). Significantly different mean values (one-way ANOVA, $p < 0.001$) between groups are indicated with “A” and “B” for organic and conventional comparison, with “a” and “b” for the geographical origin comparison and with “ α ” and “ β ” for the matrix comparison (soil, tomato, passata).

Region	System	Matrix		$\delta^2\text{H}$ (‰,vs V-SMOW)	$\delta^{18}\text{O}$ (‰,vs V-SMOW)	$\delta^{13}\text{C}$ (‰,vs V-PDB)	$\delta^{15}\text{N}$ (‰,vs AIR)	$\delta^{34}\text{S}$ (‰,vs V-CDT)
Basilicata	Conventional	Soils (n = 2)	Mean	–	–	–25.7	6.2	–
			Std.Dev.	–	–	0.4	0.4	–
		Tomatoes (n = 12)	Mean	–37	26.5	–27.9	4.6	–3.1
			Std.Dev.	8	1.1	0.3	2.7	1.3
		Passata (n = 5)	Mean	–38	24.5	–24.7	6.0	–4.4
			Std.Dev.	5	0.3	0.1	0.2	0.1
	Organic	Soils (n = 2)	Mean	–	–	–25.6	6.2	–
			Std.Dev.	–	–	0.4	0.1	–
		Tomatoes (n = 12)	Mean	–36	26.8	–27.7	6.4	–2.9
			Std.Dev.	8	0.7	0.8	1.5	1.7
		Passata (n = 3)	Mean	–38	25.0	–23.9	7.9	–4.3
			Std.Dev.	4	0.2	0.3	1.1	0.3
Emilia Romagna	Conventional	Soils (n = 2)	Mean	–	–	–24.9	6.4	–
			Std.Dev.	–	–	0.1	0.1	–
		Tomatoes (n = 12)	Mean	–37	23.4	–28.1	4.4	–0.2
			Std.Dev.	8	0.6	1.6	1.3	2.2
		Passata (n = 2)	Mean	–38	22.8	–26.1	3.2	–2.0
			Std.Dev.	1	0.2	0.2	0.8	0.3
	Organic	Soils (n = 2)	Mean	–	–	–26.7	5.2	–
			Std.Dev.	–	–	0.3	3.0	–
		Tomatoes (n = 12)	Mean	–38	25.0	–27.3	12.0	–0.5
			Std.Dev.	8	1.0	0.5	2.1	3.6
		Passata (n = 2)	Mean	–28	25.8	–23.5	13.8	1.4
			Std.Dev.	7	0.1	0.0	0.1	0.1
Region	System	Matrix		$\delta^2\text{H}$ (‰,vs V-SMOW)	$\delta^{18}\text{O}$ (‰,vs V-SMOW)	$\delta^{13}\text{C}$ (‰,vs V-PDB)	$\delta^{15}\text{N}$ (‰,vs AIR)	$\delta^{34}\text{S}$ (‰,vs V-CDT)
Basilicata and Emilia Romagna	Conventional	Soils (n = 4)	Mean	–	–	–25.3 ^A	6.3 ^A	–
			Std.Dev.	–	–	0.5	0.2	–
		Tomatoes (n = 24)	Mean	–37 ^A	25.0 ^A	–28.0 ^A	4.5 ^B	–1.7 ^A
			Std.Dev.	8	1.8	1.2	2.1	2.3
		Passata (n = 7)	Mean	–38 ^A	24.1 ^A	–25.1 ^A	5.2 ^A	–3.7 ^A
			Std.Dev.	4	0.9	0.7	1.4	1.2
	Organic	Soils (n = 4)	Mean	–	–	–26.2 ^A	5.7 ^A	–
			Std.Dev.	–	–	0.7	1.8	–
		Tomatoes (n = 24)	Mean	–37 ^A	25.9 ^A	–27.5 ^A	9.2 ^A	–1.7 ^A
			Std.Dev.	8	1.3	0.7	3.3	3.0
		Passata (n = 5)	Mean	–34 ^A	25.3 ^A	–23.7 ^A	10.2 ^A	–2.0 ^A
			Std.Dev.	7	0.5	0.4	3.4	3.1
Region	System	Matrix		$\delta^2\text{H}$ (‰,vs V-SMOW)	$\delta^{18}\text{O}$ (‰,vs V-SMOW)	$\delta^{13}\text{C}$ (‰,vs V-PDB)	$\delta^{15}\text{N}$ (‰,vs AIR)	$\delta^{34}\text{S}$ (‰,vs V-CDT)
Basilicata	Organic and conventional	Soils (n = 4)	Mean	–	–	–25.6 ^a	6.2 ^a	–
			Std.Dev.	–	–	0.3	0.2	–
		Tomatoes (n = 24)	Mean	–37 ^a	26.7 ^a	–27.8 ^a	5.5 ^a	–3.0 ^b
			Std.Dev.	8	0.9	0.6	2.3	1.5
		Passata (n = 8)	Mean	–37 ^a	24.7 ^a	–24.4 ^a	6.7 ^a	–4.4 ^b
			Std.Dev.	4	0.3	0.4	1.2	0.2
			Mean	–	–	–25.8 ^a	5.8 ^a	–
			Std.Dev.	–	–	1.1	1.8	–
Emilia Romagna	Organic and Conventional	Soils (n = 4)	Mean	–	–	–25.8 ^a	5.8 ^a	–
			Std.Dev.	–	–	1.1	1.8	–
		Tomatoes (n = 24)	Mean	–38 ^a	24.2 ^b	–27.7 ^a	8.2 ^a	–0.3 ^a
			Std.Dev.	8	1.2	1.2	4.3	2.9
		Passata (n = 4)	Mean	–33 ^a	24.3 ^a	–24.8 ^a	8.5 ^a	–0.3 ^a
			Std.Dev.	7	1.7	1.5	6.1	1.9
Region	System	Matrix		$\delta^2\text{H}$ (‰,vs V-SMOW)	$\delta^{18}\text{O}$ (‰,vs V-SMOW)	$\delta^{13}\text{C}$ (‰,vs V-PDB)	$\delta^{15}\text{N}$ (‰,vs AIR)	$\delta^{34}\text{S}$ (‰,vs V-CDT)
Basilicata and Emilia Romagna	Organic and Conventional	Soils (n = 8)	Mean	–	–	–25.7 ^{α}	6.0 ^{α}	–
			Std.Dev.	–	–	0.7	1.2	–
		Tomatoes (n = 48)	Mean	–37 ^{α}	25.4 ^{α}	–27.8 ^{β}	6.9 ^{α}	–1.7 ^{α}
			Std.Dev.	8	1.6	1.0	3.7	2.7
		Passata (n = 12)	Mean	–36 ^{α}	24.6 ^{α}	–24.5 ^{α}	7.3 ^{α}	–3.0 ^{α}
			Std.Dev.	6	1.0	0.9	3.5	2.2

Table 2
Amino acid $\delta^{15}\text{N}_{\text{aa}}$ and $\delta^{13}\text{C}_{\text{aa}}$ values determined on a subset of tomato samples ($N = 16$) grown organically and conventionally in 2012 and 2013 in two Italian regions (Emilia Romagna and Basilicata). Significantly different mean values (one-way ANOVA, $p < 0.001$) between groups are indicated with “a” and “b” for organic and conventional comparison, with “a” and “b” for the geographical origin comparison and with “c” and “d” for the year comparison.

	$\delta^{13}\text{C}_{\text{Bulk}}$	$\delta^{13}\text{C}_{\text{Ala}}$	$\delta^{13}\text{C}_{\text{Val}}$	$\delta^{13}\text{C}_{\text{Leu}}$	$\delta^{13}\text{C}_{\text{Leu}}$	$\delta^{13}\text{C}_{\text{Gly}}$	$\delta^{13}\text{C}_{\text{Pro}}$	$\delta^{13}\text{C}_{\text{Thr}}$	$\delta^{13}\text{C}_{\text{Glx}}$	$\delta^{13}\text{C}_{\text{Phe}}$	$\delta^{15}\text{N}_{\text{Bulk}}$	$\delta^{15}\text{N}_{\text{Ala}}$	$\delta^{15}\text{N}_{\text{Val}}$	$\delta^{15}\text{N}_{\text{Ileu}}$	$\delta^{15}\text{N}_{\text{Leu}}$	$\delta^{15}\text{N}_{\text{Gly}}$	$\delta^{15}\text{N}_{\text{Pro}}$	$\delta^{15}\text{N}_{\text{Thr}}$	$\delta^{15}\text{N}_{\text{Glx}}$	$\delta^{15}\text{N}_{\text{Phe}}$
Organic (N = 8)	Mean	-27.4 ^A	-27.3 ^A	-33.8 ^A	-25.5 ^A	-36.5 ^A	-25.8 ^A	-22.7 ^A	-26.9 ^A	-29.2 ^A	9.0 ^A	6.9 ^A	8.5 ^A	5.6 ^A	2.9 ^A	8.3 ^A	15.6 ^A	6.3 ^A	10.1 ^A	12.9 ^A
	Std.Dev.	0.8	0.8	0.8	1.2	0.9	1.7	2.2	0.6	0.7	3.8	3.3	3.5	3.0	3.3	3.7	3.4	3.3	2.6	2.8
Conventional (N = 8)	Mean	-28.0 ^A	-27.8 ^A	-34.0 ^A	-25.6 ^A	-36.9 ^A	-26.2 ^A	-23.7 ^A	-28.4 ^B	-29.4 ^A	4.8 ^A	1.9 ^A	2.8 ^A	-0.2 ^B	-2.5 ^A	3.7 ^A	10.9 ^A	1.0 ^A	4.4 ^B	7.8 ^B
	Std.Dev.	1.2	1.2	0.7	1.3	0.9	2.3	2.5	0.5	0.9	2.5	2.5	2.1	2.0	2.3	3.5	2.0	2.2	2.2	1.6
Basilicata (N = 8)	Mean	-27.7 ^a	-27.7 ^a	-34.2 ^a	-25.8 ^a	-37.1 ^a	-26.0 ^a	-23.6 ^a	-27.8 ^a	-29.8 ^a	5.4 ^a	2.8 ^a	4.2 ^a	1.1 ^a	-1.3 ^a	4.4 ^a	11.9 ^a	2.7 ^a	6.1 ^a	9.1 ^a
	Std.Dev.	0.8	0.8	0.7	0.6	0.5	1.2	1.7	0.9	0.6	2.4	3.3	2.1	2.3	2.1	3.6	2.4	1.8	3.1	2.6
Emilia Romagna (N = 8)	Mean	-27.8 ^a	-27.4 ^b	-33.6 ^a	-25.3 ^a	-36.3 ^a	-26.0 ^a	-22.7 ^a	-27.6 ^a	-28.8 ^a	8.3 ^a	5.9 ^a	7.1 ^a	4.3 ^a	1.7 ^a	7.6 ^a	14.7 ^a	4.6 ^a	8.4 ^a	11.5 ^a
	Std.Dev.	1.3	1.2	0.6	1.6	1.1	2.5	2.8	1.0	0.6	4.5	3.8	5.1	4.5	4.9	4.3	4.5	5.0	4.2	3.9
2012 (N = 8)	Mean	-27.5 ^a	-27.3 ^a	-33.6 ^a	-25.2 ^a	-36.6 ^a	-25.1 ^a	-22.4 ^a	-27.5 ^a	-29.1 ^a	6.1 ^a	3.4 ^a	5.7 ^a	2.9 ^a	0.3 ^a	4.7 ^a	12.4 ^a	3.5 ^a	6.6 ^a	10.5 ^a
	Std.Dev.	0.7	0.9	0.6	1.1	0.8	1.4	2.0	0.9	0.8	3.2	3.5	3.6	3.7	3.6	3.9	3.3	3.3	3.5	3.5
2013 (N = 8)	Mean	-27.9 ^a	-27.8 ^a	-34.2 ^a	-25.9 ^a	-36.8 ^a	-26.9 ^a	-23.9 ^a	-27.9 ^a	-29.5 ^a	7.7 ^a	5.3 ^a	5.6 ^a	2.5 ^a	0.1 ^a	7.2 ^a	14.2 ^a	3.9 ^a	8.0 ^a	10.1 ^a
	Std.Dev.	1.2	1.1	0.7	1.2	1.1	2.3	2.3	0.9	0.7	4.4	4.1	4.7	4.3	4.5	4.4	4.2	4.4	4.1	3.6

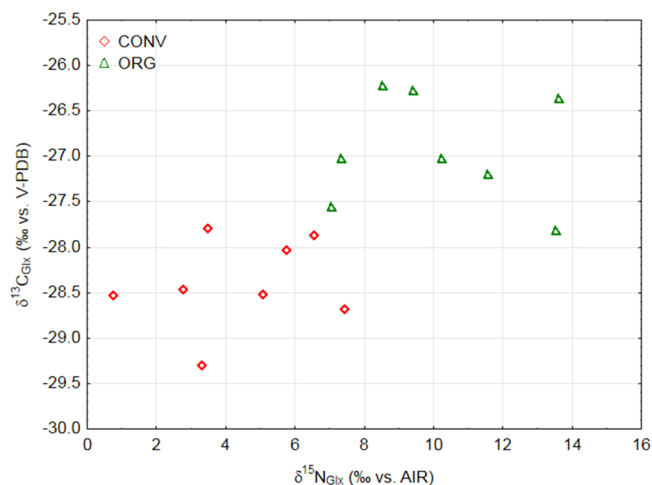


Fig. 2. Tomato $\delta^{15}\text{N}_{\text{Glx}}$ samples plotted against $\delta^{13}\text{C}_{\text{Glx}}$ samples. Org (organic), Conv (conventional).

2014b). The amino acids are all derived from glutamine and glutamic acid, therefore the single amino acid isotopic values were normalized against the corresponding glutamine + glutamate values (Glx) to cancel out the differences due to peripheral sources of nitrogen other than the agricultural system (*i.e.* the release of ammonia and/or nitrate during organic matter decomposition or the conversion of atmospheric nitrogen into these compounds by natural processes, such as lightning, and biological nitrogen fixation by bacteria). The $\delta^{15}\text{N}$ values, for both farming systems, of Pro ($6.0\text{‰} \pm 2.1\text{‰}$) and Phe ($3.0\text{‰} \pm 1.6\text{‰}$) were higher when compared to the $\delta^{15}\text{N}$ values for Glx. Phe is enriched in $\delta^{15}\text{N}$ compared to Glx due to a kinetic isotope effect in the synthesis of phenylpropanoids (for structural and protection purposes in plants) where it is deaminated as an intermediate (Styring et al., 2014b; Werner & Schmidt, 2002). Pro is also enriched in $\delta^{15}\text{N}$ compared to Glx and this is due to a kinetic isotope effect associated with its catabolism via enzymatic reactions (Hermes, Weiss, & Cleland, 1985; Styring et al., 2014b). The $\delta^{15}\text{N}$ values for Ala ($-2.9\text{‰} \pm 1.3\text{‰}$), Ileu ($-4.6\text{‰} \pm 1.7\text{‰}$) and Leu ($-7.1\text{‰} \pm 1.6\text{‰}$) were lower when compared to the $\delta^{15}\text{N}$ values for Glx. Similarly, the majority of the values of Val ($-1.6\text{‰} \pm 1.7\text{‰}$) Gly ($-1.3\text{‰} \pm 1.9\text{‰}$) and Thr ($-3.6\text{‰} \pm 2.1\text{‰}$) were lower when compared to the $\delta^{15}\text{N}$ values for Glx. Val, Ileu and Leu are branch-chain amino acids (BCAA) and are precursors to synthesizing proteins (Styring et al., 2014b; Werner & Schmidt, 2002). Val is synthesized directly from pyruvate, which is derived from photosynthesis (Bryan, 1980) and Leu is synthesized from the Val metabolic pathway via a succession of reactions branching off from it (Kochovenko & Fernie, 2011; Pátek, 2007) and these successive reactions could be the cause of the greater discrimination for $\delta^{15}\text{N}$ in Leu than for Val (Bol, Ostle, & Petzke, 2002; Hofmann, Jung, Segschneider, Gehre, & Schüürmann, 1995). Ileu is synthesized from Thr, which is deaminated, and during this metabolic process, there is a slight depletion in $\delta^{15}\text{N}$ for Ileu compared to Thr. Gly is involved in the photorespiratory nitrogen cycle and is derived from the Glu, Ala or Asp amino group (Styring et al., 2014b; Werner & Schmidt, 2002). Gly $\delta^{15}\text{N}$ is enriched compared with Ala and is slightly depleted to Glx. Ala is also slightly depleted compared to Glx.

Also for $\delta^{13}\text{C}$, to remove the external factors that could be affecting the isotopic value such as the atmosphere, the amino acid $\delta^{13}\text{C}$ values were normalized to the Glx $\delta^{13}\text{C}$ value (Fig. 2 Supplementary Material). The $\delta^{13}\text{C}$ values for both farming systems and regions of Pro ($1.7\text{‰} \pm 2.1\text{‰}$), Ileu ($2.1\text{‰} \pm 1.1\text{‰}$) and Thr ($4.5\text{‰} \pm 2.1\text{‰}$) are higher when compared to the $\delta^{13}\text{C}$ values for Glx. The $\delta^{13}\text{C}$ values for Ala ($0.1\text{‰} \pm 1.0\text{‰}$) are almost always very similar as for Glx. The $\delta^{13}\text{C}$ values for Val ($-6.2\text{‰} \pm 1.0\text{‰}$), Leu ($-9.0\text{‰} \pm 0.8\text{‰}$), Gly ($-11.2\text{‰} \pm 1.9\text{‰}$) and Phe ($-1.6\text{‰} \pm 1.0\text{‰}$) were lower when

Table 3
Bulk $\delta^2\text{H}$, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, $\delta^{18}\text{O}$, $\delta^{34}\text{S}$ and amino acid $\delta^{15}\text{N}_{\text{aa}}$ and $\delta^{13}\text{C}_{\text{aa}}$ values determined on commercial Danish tomato samples (N = 13) organically and conventionally grown.

Number	System	Media	Fertilizer	$\delta^2\text{H}_{\text{bulk}}$	$\delta^{13}\text{C}_{\text{bulk}}$	$\delta^{15}\text{N}_{\text{bulk}}$	$\delta^{18}\text{O}_{\text{bulk}}$	$\delta^{34}\text{S}_{\text{bulk}}$	$\delta^{13}\text{C}_{\text{Ala}}$	$\delta^{13}\text{C}_{\text{Val}}$	$\delta^{13}\text{C}_{\text{Ileu}}$	$\delta^{15}\text{N}_{\text{Glx}}$	$\delta^{13}\text{C}_{\text{Glu}}$	$\delta^{15}\text{N}_{\text{Phe}}$
1	Organic	Soil	Manure, liquid	-65	-29.5	3.9	23.7	0.8	-26.4	-32.3	-24.9	-36.0	-38.7	
2	Organic	Soil	Manure, liquid	-55	-29.6	4.7	25.3	1.2	-29.3	-36.0	-27.6	-37.9	-40.7	
3	Organic	Peat	Liquid organic	-69	-39.5	1.1	23.5	-0.6	-38.5	-44.9	-37.1	-46.6	-51.0	
4	Organic	Peat	Liquid organic	-53	-43.5	1.6	24.6	-0.8	-44.2	-49.4	-41.7	-50.8	-61.0	
5	Organic	Peat	Liquid organic	-61	-44.3	0.8	24.9	-1.3	-42.0	-45.7	-38.8	-49.0	-55.7	
6	Organic	Soil	Manure	-54	-31.3	10.5	23.3	2.3	-29.7	-36.7	-27.5	-38.6	-42.4	
7	Organic	Soil	Manure	-42	-32.6	8.6	24.3	2.5	-30.0	-35.0	-27.8	-38.6	-45.0	
8	Organic	Soil	Manure	-56	-32.5	10.5	23.4	2.3	-30.2	-34.8	-26.8	-37.1	-44.0	
9	Conventional	Inert	Liquid inorganic	-65	-33.6	-0.1	23.8	-0.7	-32.8	-38.8	-30.2	-40.7	-44.6	
10	Conventional	Inert	Liquid inorganic	-65	-39.4	1.8	23.2	1.6	-38.4	-45.2	-37.2	-47.3	-48.5	
11	Conventional	Inert	Liquid inorganic	-51	-35.9	1.3	24.1	0.8	-33.5	-37.0	-31.6	-42.2	-49.9	
12	Conventional	Inert	Liquid inorganic	-55	-37.9	2.2	23.3	4.7	-34.6	-41.2	-36.8	-46.5	-47.2	
13	Conventional	Inert	Liquid inorganic	-56	-38.2	2.3	24.9	5.7	-36.1	-44.1	-34.5	-45.1	-45.4	
	Organic	Mean		-57	-35.4	5.2	24.1	0.8	-33.8	-39.3	-31.5	-41.8	-47.3	
		Std. Dev.		8	6.1	4.1	0.8	1.5	6.7	6.3	6.5	5.9	7.8	
	Conventional	Mean		-58	-37.0	1.5	23.9	2.4	-35.1	-41.2	-34.0	-44.4	-47.1	
		Std. Dev.		6	2.3	1.0	0.7	2.7	2.2	3.5	3.1	2.8	2.2	

Number	$\delta^{13}\text{C}_{\text{Pro}}$	$\delta^{13}\text{C}_{\text{Thr}}$	$\delta^{13}\text{C}_{\text{Glx}}$	$\delta^{13}\text{C}_{\text{Phe}}$	$\delta^{15}\text{N}_{\text{Ala}}$	$\delta^{15}\text{N}_{\text{Val}}$	$\delta^{15}\text{N}_{\text{Ileu}}$	$\delta^{15}\text{N}_{\text{Leu}}$	$\delta^{15}\text{N}_{\text{Gly}}$	$\delta^{15}\text{N}_{\text{Pro}}$	$\delta^{15}\text{N}_{\text{Thr}}$	$\delta^{15}\text{N}_{\text{Glx}}$	$\delta^{15}\text{N}_{\text{Phe}}$
1	-26.7	-20.9	-26.5	-30.0	2.3	3.8	-0.9	-4.5	6.2	4.6	2.9	4.6	5.9
2	-26.8	-24.8	-26.7	-29.2	2.7	3.3	-0.2	-1.9	4.1	9.5	2.5	4.9	7.9
3	-37.0	-36.4	-39.6	-38.4	-2.2	-1.6	-4.8	-6.9	1.9	5.7	-3.0	1.4	4.0
4	-42.1	-39.6	-45.9	-41.3	-1.3	-0.3	-3.8	-4.4	2.2	3.5	-1.9	3.2	6.3
5	-39.5	-39.3	-43.2	-39.7	-0.5	0.3	-2.3	-3.9	1.9	2.4	-1.4	2.0	5.1
6	-28.8	-23.3	-29.9	-31.0	6.9	8.1	6.3	5.8	8.8	13.0	11.1	10.5	12.1
7	-28.8	-26.2	-31.6	-31.1	6.1	6.1	4.5	3.9	8.7	8.6	4.5	9.7	13.1
8	-28.3	-23.2	-31.6	-30.2	8.2	8.9	7.3	7.1	8.3	12.1	6.1	11.6	14.7
9	-30.3	-25.9	-32.3	-32.3	-1.2	-1.6	-4.9	-4.4	0.5	2.8	-6.6	1.1	4.8
10	-37.4	-35.9	-39.3	-38.0	0.0	0.1	-3.8	-3.2	2.6	3.5	-0.8	2.9	4.3
11	-33.2	-26.7	-36.0	-34.1	-2.2	-0.9	-3.6	-3.1	0.2	1.5	-6.2	2.6	4.8
12	-36.6	-36.1	-37.4	-37.2	-1.4	0.5	-2.3	-2.2	-0.2	5.7	-2.1	2.9	4.6
13	-33.8	-32.2	-36.3	-37.2	-0.4	-0.6	-2.0	-1.5	1.8	6.2	1.0	3.3	4.7
	-32.3	-29.2	-34.4	-33.9	2.8	3.6	0.8	-0.6	5.3	7.4	2.6	6.0	8.6
	6.3	7.8	7.5	5.0	4.0	3.9	4.7	5.4	3.1	4.0	4.7	4.0	4.1
	-34.3	-31.4	-36.3	-35.8	-1.0	-0.5	-3.3	-2.9	1.0	3.9	-3.0	2.6	4.7
	2.8	4.9	2.6	2.5	0.9	0.8	1.2	1.1	1.2	2.0	3.4	0.8	0.2

compared to the $\delta^{13}\text{C}$ values for Glx.

The fixation of CO_2 in plants by the enzyme RuBisCO has a kinetic isotope effect. RuBisCO generates glucose and aids in the photorespiration of the plant; these affect the amino acids involved in these systems, such as glycine in photorespiration (Paolini et al., 2015; Styring et al., 2014b; Werner & Schmidt, 2002). Val is derived from photosynthesis as is Leu, by subsequent branching reactions from the Val metabolic pathway. Gly is derived from photosynthesis and this can explain the lower $\delta^{13}\text{C}$ values for these 3 amino acids: Val (-35.3‰ to -32.7‰), Leu (-38.0‰ to -34.8‰) and Gly (-44.2‰ to -36.7‰) (Table 2).

A separation between the farming systems, organic and conventional, as observed when $\delta^{13}\text{C}_{\text{Glx}}$ values were plotted against $\delta^{15}\text{N}_{\text{Glx}}$ values (Fig. 2). A clear improvement of the discrimination capacity was evident when compared with the plot of $\delta^{13}\text{C}_{\text{bulk}}$ and $\delta^{15}\text{N}_{\text{bulk}}$ determined for the same 16 samples (Fig. 3 Supplementary Material). This demonstrated that compound-specific isotope analysis provided added classification power for organic authentication compared to traditional bulk tissue isotope analysis.

Principal component analysis was conducted using the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of the amino acids that showed statistically significant differences at ANOVA test for $p < 0.001$ ($\delta^{15}\text{N}$ values of Ileu, Glx and Phe and $\delta^{13}\text{C}_{\text{Glx}}$) (Fig. 4 Supplementary Material). The majority of the data variability was explained with the first two PCs with 86% variance explained for the first PC and 10% for the second PC. The first PC had a number of amino acid contributors: all the $\delta^{15}\text{N}$ values of Ileu, Glx and Phe that were highly correlated. The major contributor of PC2 was Glx for $\delta^{13}\text{C}$. Analyzing the PCA for farming system for amino acid $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values (Fig. 4 Supplementary Material) a much better separation between the organic and conventional systems was obtained compared to the bulk tomato data. The majority of the organically grown samples had higher values of $\delta^{15}\text{N}_{\text{aa}}$ and $\delta^{13}\text{C}_{\text{Glx}}$ compared to the conventionally grown ones. This corresponds with the findings of Paolini and colleagues (Paolini et al., 2015) for wheat samples.

To further test the efficacy of the isotopic profile of amino acids for the differentiation of organic and conventional tomatoes, the method was applied to 13 tomato samples collected from Danish growers (Table 3). Samples were initially analyzed for stable isotope ratios of H, C, N, O and S, on bulk tissue. The ranges of values for the five stable isotope ratios were overlapped between the conventional and organic products, thus not allowing for discrimination of these samples. In particular, three organic samples showed low values of $\delta^{15}\text{N}$ probably due to the use of organic liquid fertilizers derived from nitrogen fixing plants. Furthermore, it should be noted that the test samples had very low values of $\delta^{13}\text{C}$ (both bulk and amino acids). Schmidt and colleagues (2005) suggested that in German tomatoes $\delta^{13}\text{C}_{\text{bulk}}$ lower than -28‰ indicated greenhouse-cultivated tomatoes with methane based heating. Data of $\delta^{13}\text{C}_{\text{Glx}}$ was plotted against $\delta^{15}\text{N}_{\text{Glx}}$ values to visualize separation between the farming systems; organic and conventional (Fig. 5 Supplementary Material). Due to the lower values of $\delta^{13}\text{C}$ from the greenhouse grown tomatoes, it was decided to normalize $\delta^{13}\text{C}_{\text{Glx}}$ values against $\delta^{13}\text{C}_{\text{bulk}}$ and plot the normalized data against $\delta^{15}\text{N}_{\text{Glx}}$ (Fig. 3). The conventional and organic samples of the test set grouped in two different parts of the plot similarly to the organic and conventional samples of the subset of 16 Italian field-grown tomatoes, with the exception of three organic test samples that fell in the conventional group. These samples were found to have comparably lower values of $\delta^{15}\text{N}_{\text{bulk}}$ and $\delta^{15}\text{N}_{\text{aa}}$ (Table 3). An explanation of this could be that these three samples were the only ones grown on peat and that the fertilization strategy was partly based on legumes. The stable isotopic composition of peat-forming plants, including $\delta^{15}\text{N}$, is well preserved in a peat formed in cold regions (Skrzypek, Paul, & Wojtuń, 2008) and we propose that the peat used as media in this study originated from nitrogen fixing plants.

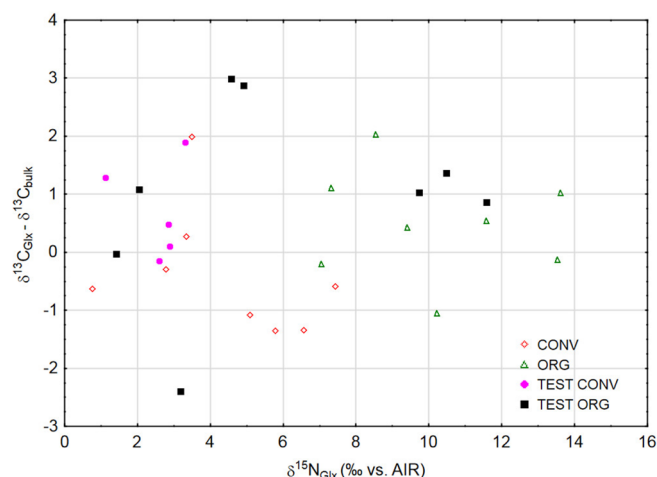


Fig. 3. $\delta^{15}\text{N}_{\text{Glx}}$ and $\delta^{13}\text{C}_{\text{Glx}}$ values normalized according to $\delta^{15}\text{N}_{\text{bulk}}$ and $\delta^{13}\text{C}_{\text{bulk}}$ plotted against each other for tomato samples of model and test samples. Org (organic), Conv (conventional).

4. Conclusions

In this study, the isotopic signatures ($\delta^2\text{H}$, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, $\delta^{18}\text{O}$ and $\delta^{34}\text{S}$) of soils, tomatoes and tomato passata samples collected from two Italian regions, produced organically or conventionally in two years, were investigated along all steps of the production chain of passata. The main factors influencing the bulk isotopic signatures of tomatoes and passata were the geographical origin and the agricultural production system. Differences between conventional and organic agricultural practices were primarily detected for $\delta^{15}\text{N}$, as expected, even if this parameter alone is not able to completely distinguish organic and conventionally grown products.

By analyzing stable isotope ratios in amino acids from field-grown tomatoes (Ala, Val, Ileu, Leu, Gly, Pro, Thr, Glx and Phe), it was possible to improve the discrimination power according to the production system. In particular, $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ of Glx provided added value for organic authentication compared to isotopic bulk data. However, when applying the compound-specific analysis to a test set of greenhouse-grown tomatoes, the discrimination of organic and conventional tomatoes was less successful. In particular, $\delta^{13}\text{C}_{\text{Glx}}$ was biased by greenhouse growth conditions likely due to a higher CO_2 concentration and air humidity compared to field growth conditions. This clearly demonstrates that the growth conditions (greenhouse or field) should be taken into consideration when authenticating organic produce. Overall, it can be concluded that compound-specific isotopic analysis was superior for authentication of organic, field-grown tomatoes and efficiently eliminated the limitations of bulk isotope analysis. However, for greenhouse-grown tomatoes further studies should be conducted to investigate the applicability of amino acid isotope analysis for authentication of organically grown produce. Furthermore, a wider sampling plan and complementary analytical techniques including pesticide residue analysis and isotope ratio analysis of oxy-anions such as nitrate and sulfate, are also advisable in order to improve the discrimination power and the applicability of the developed method for authentication of organically grown produce.

CRedit authorship contribution statement

Luana Bontempo: Conceptualization, Methodology, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Katryna A. van Leeuwen:** Methodology, Writing - original draft, Writing - review & editing. **Mauro Paolini:** Methodology, Validation, Writing - original draft. **Kristian Holst Laursen:** Conceptualization, Writing - original draft, Writing - review & editing. **Cristina Micheloni:**

Writing - original draft, Writing - review & editing. **Paul D. Prenzler:** Writing - original draft, Writing - review & editing. **Danielle Ryan:** Writing - original draft, Writing - review & editing. **Federica Camin:** Conceptualization, Methodology, Validation, Writing - original draft, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2020.126426>.

References

- Bateman, A. S., & Kelly, S. D. (2007). Fertilizer nitrogen isotope signatures. *Isotopes in Environmental and Health Studies*, 43(3), 237–247.
- Bol, R., Ostle, N. J., & Petzke, K. J. (2002). Compound specific plant amino acid $\delta^{15}\text{N}$ values differ with functional plant strategies in temperate grassland. *Journal of Plant Nutrition and Soil Science*, 165(6), 661–667.
- Bontempo, L., Camin, F., Manzocco, L., Nicolini, G., Wehrens, R., Ziller, L., & Larcher, R. (2011). Traceability along the production chain of Italian tomato products on the basis of stable isotopes and mineral composition. *Rapid Communications in Mass Spectrometry: RCM*, 25(7), 899–909.
- Bryan, J. K. (1980). Synthesis of the Aspartate Family and Branched-Chain Amino Acids. In *Amino Acids and Derivatives* (pp. 403–452).
- Camin, F., Perini, M., Bontempo, L., Fabroni, S., Faedi, W., Magnani, S., ... Rapisarda, P. (2011). Potential isotopic and chemical markers for characterising organic fruits. *Food Chemistry*, 125(3), 1072–1082.
- Cristea, G., Feher, I., Magdas, D. A., Voica, C., & Puscas, R. (2017). Characterization of vegetables by stable isotopic and elemental signatures. *Analytical Letters*, 50(17), 2677–2690.
- Dehelean, A., Magdas, D. A., & Cristea, G. (2013). Investigation of trace metals content and carbon isotopic composition on the soil leaf-fruit chain from some transylvanian areas. *Analytical Letters*, 46(3), 498–507.
- Docherty, G., Jones, V., & Evershed, R. P. (2001). Practical and theoretical considerations in the gas chromatography/combustion/isotope ratio mass spectrometry d^{13}C analysis of small polyfunctional compounds. *Rapid Communications in Mass Spectrometry: RCM*, 15(9), 730–738.
- Dunn, P. J. H., & Carter, J. F. (2018). Publication of the second edition of the FIRMS good practice guide for Isotope Ratio Mass Spectrometry. *Rapid Communications in Mass Spectrometry: RCM*. <https://doi.org/10.1002/rcm.8299>.
- Flores, P., López, A., Fenoll, J., Hellín, P., & Kelly, S. (2013). Classification of organic and conventional sweet peppers and lettuce using a combination of isotopic and biomarkers with multivariate analysis. *Journal of Food Composition and Analysis: An Official Publication of the United Nations University, International Network of Food Data Systems*, 31(2), 217–225.
- Georgi, M., Voerkelius, S., Rossmann, A., Graßmann, J., & Schnitzler, W. H. (2005). Multiple isotope ratios of vegetables from integrated and organic production. *Plant and Soil*, 275(1–2), 93–100.
- Hayes, W. A., Smith, P. G., & Morris, A. E. (1998). The production and quality of tomato concentrates. *Critical Reviews in Food Science and Nutrition*, 38(7), 557–564.
- Hermes, J. D., Weiss, P. M., & Cleland, W. W. (1985). Use of nitrogen-15 and deuterium isotope effects to determine the chemical mechanism of phenylalanine ammonia-lyase. *Biochemistry*, 24(12), 2959–2967.
- Hofmann, D., Jung, K., Segschneider, J. H., Gehre, M., & Schüürmann, G. (1995). $^{15}\text{N}/^{14}\text{N}$ analysis of amino acids with GC-C-IRMS – Methodical investigations and ecotoxicological applications. *Isotopes in Environmental and Health Studies*, 31(3–4), 367–375.
- Hong, E., Lee, S. Y., Jeong, J. Y., Park, J. M., Kim, B. H., Kwon, K., & Chun, H. S. (2017). Modern analytical methods for the detection of food fraud and adulteration by food category. *Journal of the Science of Food and Agriculture*, 97(12), 3877–3896.
- Kahl, J., Bodroza-Solarov, M., Busscher, N., Hajslova, J., Kneifel, W., Kokornaczky, M. O., ... Stolz, P. (2014). Status quo and future research challenges on organic food quality determination with focus on laboratory methods. *Journal of the Science of Food and Agriculture*, 94(13), 2595–2599.
- Kelly, S. D., Mihailova, A. (2017). Chapter 12: Organic food authenticity. J.F. Carter, L.A. Chesson (Eds.), *Food forensics: Stable isotopes as a guide to authenticity and origin*, 273–302.
- Kochevenko, A., & Fernie, A. R. (2011). The genetic architecture of branched-chain amino acid accumulation in tomato fruits. *Journal of Experimental Botany*, 62(11), 3895–3906.
- Krouse, H. R., & Grinenko, V. A. (1991). International Council of Scientific Unions. Scientific Committee on Problems of the Environment. Stable Isotopes: Natural and Anthropogenic Sulphur in the Environment. John Wiley & Son Limited.
- Krueger, H. W., & Reesman, R. H. (1982). Carbon isotope analyses in food technology. *Mass Spectrometry Reviews*, 1(3), 205–236.
- Laursen, K. H., Mihailova, A., Kelly, S. D., Epov, V. N., Bérail, S., Schjoerring, J. K., ... Husted, S. (2013). Is it really organic? – Multi-isotopic analysis as a tool to discriminate between organic and conventional plants. *Food Chemistry*, 141(3), 2812–2820.
- Laursen, K. H., Schjoerring, J. K., Kelly, S. D., & Husted, S. (2014). Authentication of organically grown plants – advantages and limitations of atomic spectroscopy for multi-element and stable isotope analysis. *Trends in Analytical Chemistry: TRAC*, 59, 73–82.
- Lea, P. J., & Azevedo, R. A. (2017). Amino Acids. In *Encyclopedia of Applied Plant Sciences* (pp. 56–66).
- Longobardi, F., Innamorato, V., Di Gioia, A., Ventrella, A., Lippolis, V., Logrieco, A. F., ... Agostiano, A. (2017). Geographical origin discrimination of lentils (*Lens culinaris* Medik.) using H NMR fingerprinting and multivariate statistical analyses. *Food Chemistry*, 237, 743–748.
- Mahne Opatić, A., Opatić, A. M., Nečemer, M., Kocman, D., & Lojen, S. (2017). Geographical origin characterization of slovenian organic garlic using stable isotope and elemental composition analyses. *Acta Chimica Slovenica*, 1048–1055.
- Masclaux-Daubresse, C., Daniel-Vedele, F., Dechorgnat, J., Chardon, F., Gaufichon, L., & Suzuki, A. (2010). Nitrogen uptake, assimilation and remobilization in plants: Challenges for sustainable and productive agriculture. *Annals of Botany*, 105(7), 1141–1157.
- Mihailova, A., Pedentchouk, N., & Kelly, S. D. (2014). Stable isotope analysis of plant-derived nitrate – novel method for discrimination between organically and conventionally grown vegetables. *Food Chemistry*, 154, 238–245.
- Mimmo, T., Camin, F., Bontempo, L., Capici, C., Tagliavini, M., Cesco, S., & Scampicchio, M. (2015). Traceability of different apple varieties by multivariate analysis of isotope ratio mass spectrometry data. *Rapid Communications in Mass Spectrometry: RCM*, 29(21), 1984–1990.
- Nakano, A., & Uehara, Y. (2007). Effects of different kinds of fertilizer and application methods on d^{15}N values of tomato. *Japan Agricultural Research Quarterly: JARQ*, 41(3), 219–226.
- Novak, V., Adler, J., Husted, S., Fromberg, A., & Laursen, K. H. (2019). Authenticity testing of organically grown vegetables by stable isotope ratio analysis of oxygen in plant-derived sulphate. *Food Chemistry*, 291, 59–67.
- O'Leary, M. H. (1981). Carbon isotope fractionation in plants. *Phytochemistry*, 20(4), 553–567.
- Paolini, M., Ziller, L., Laursen, K. H., Husted, S., & Camin, F. (2015). Compound-specific $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ analyses of amino acids for potential discrimination between organically and conventionally grown wheat. *Journal of Agricultural and Food Chemistry*, 63(25), 5841–5850.
- Pátek, M. (2007). Branched-Chain Amino Acids. In *Microbiology Monographs* (pp. 129–162).
- Rapisarda, P., Camin, F., Fabroni, S., Perini, M., Torrisi, B., & Intrigliolo, F. (2010). Influence of different organic fertilizers on quality parameters and the $\delta^{15}\text{N}$, $\delta^{13}\text{C}$, $\delta^{2}\text{H}$, $\delta^{34}\text{S}$, and $\delta^{18}\text{O}$ values of orange fruit (*Citrus sinensis* L. Osbeck). *Journal of Agricultural and Food Chemistry*, 58(6), 3502–3506.
- Schmidt, H.-L., Rossmann, A., Voerkelius, S., Schnitzler, W. H., Georgi, M., Grassmann, J., ... Winkler, R. (2005). Isotope characteristics of vegetables and wheat from conventional and organic production. *Isotopes in Environmental and Health Studies*, 41(3), 223–228.
- Skrzypek, G., Paul, D., & Wojtuś, B. (2008). Stable isotope composition of plants and peat from Arctic mire and geothermal area in Iceland. *Polish Polar Research*, 29(4), 365–376.
- Styring, A. K., Fraser, R. A., Bogaard, A., & Evershed, R. P. (2014a). Cereal grain, rachis and pulse seed amino acid $\delta^{15}\text{N}$ values as indicators of plant nitrogen metabolism. *Phytochemistry*, 97, 20–29.
- Styring, A. K., Fraser, R. A., Bogaard, A., & Evershed, R. P. (2014b). The effect of manuring on cereal and pulse amino acid $\delta^{15}\text{N}$ values. *Phytochemistry*, 102, 40–45.
- Styring, A. K., Kuhl, A., Knowles, T. D. J., Fraser, R. A., Bogaard, A., & Evershed, R. P. (2012). Practical considerations in the determination of compound-specific amino

- acid $\delta^{15}\text{N}$ values in animal and plant tissues by gas chromatography-combustion-isotope ratio mass spectrometry, following derivatisation to their N-acetylisopropyl esters. *Rapid Communications in Mass Spectrometry: RCM*, 26(19), 2328–2334.
- van Leeuwen, K. A., Prenzler, P. D., Ryan, D., & Camin, F. (2014). Gas Chromatography-Combustion-Isotope Ratio Mass Spectrometry for traceability and authenticity in foods and beverages. *Comprehensive Reviews in Food Science and Food Safety*, 13(5), 814–837.
- Wada, E. (1995). Stable isotopes in the biosphere.
- Werner, R. A., & Schmidt, H.-L. (2002). The in vivo nitrogen isotope discrimination among organic plant compounds. *Phytochemistry*, 61(5), 465–484.
- Yarnes, C. T., & Herszage, J. (2017). The relative influence of derivatization and normalization procedures on the compound-specific stable isotope analysis of nitrogen in amino acids. *Rapid Communications in Mass Spectrometry: RCM*, 31(8), 693–704.